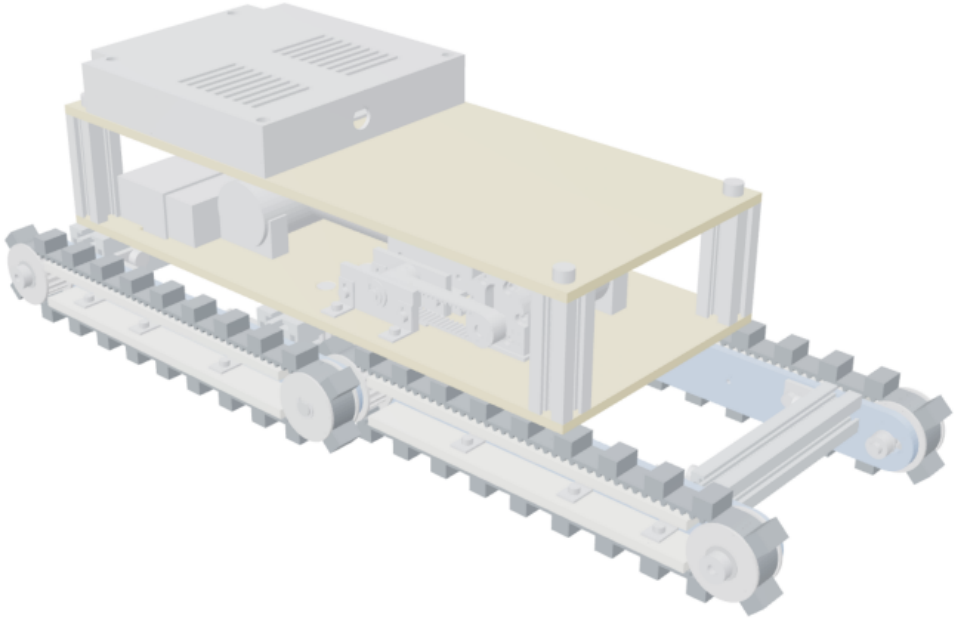
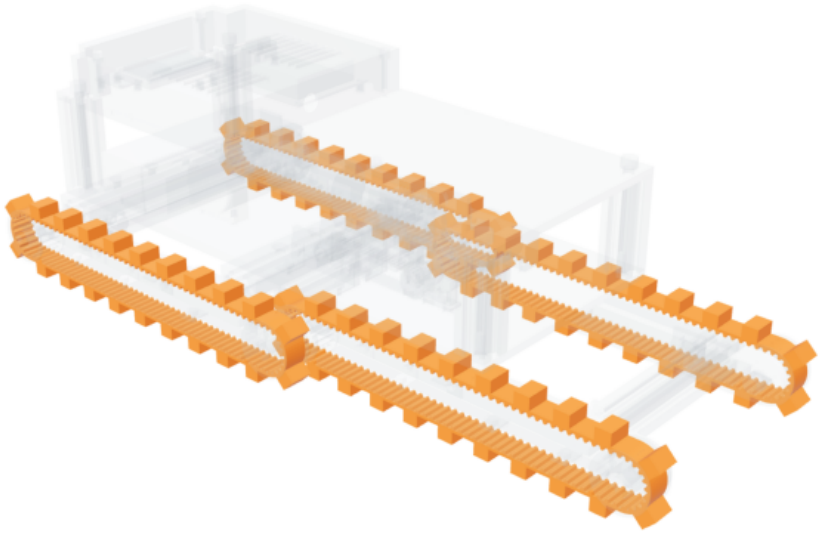
Parts catalogue, hybrid: my track on Nhien's Rev B frame



One page per part. Orange = that part in Nhien's Rev B assembly (their CAD, 347 parts). Blue-ish parts are links: click the vendor line to open the product page. Rev B sheet numbers refer to 'RB4107 Fabrication drawings (Rev B).pdf'.

Mine to make: track x4.
Mine to buy: pulleys x8, TPU.
Nhien's: plates, rails, shafts, bearings, shoulder screws, bars, deck, motors, electronics (their BOM).

Track



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 4

MAKE: 3D print (mine)

What it does

The printed TPU belt: 110 teeth inside on the toothed wheels, 45 cleats outside matched to the ES1 step. Unchanged by the plate swap; fits Rev B's pulleys and 225 mm spacing. Rev B's own alternative is a stock 550-5M-15 belt with 22 glued cleats.

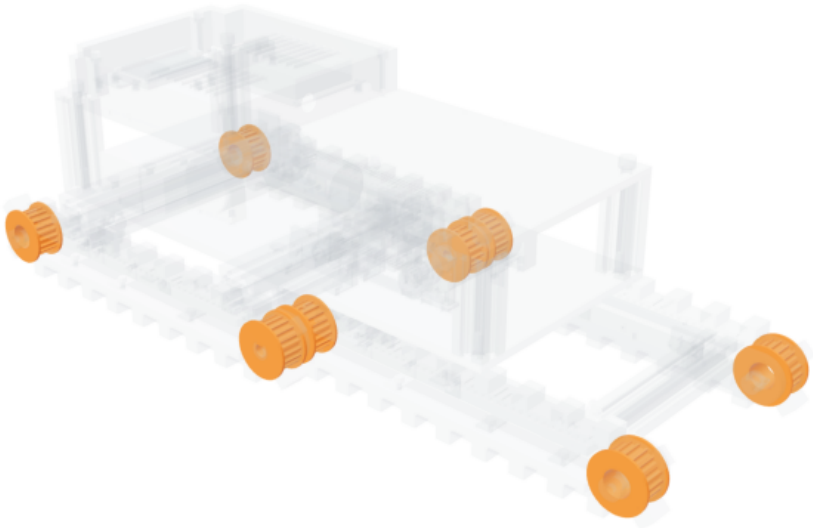
Where to get it

Fab Lab E4-02-06, Tiertime UP300 with the TPU head. File stl/track_TPU_x4.stl, standing as a ring, no supports, 0.2 mm, 100% infill. Filament: Fab Lab stock or Element14 SG TPU 95A (link).

Link: Element14 SG: TPU 95A filament

https://sg.element14.com/multicomp-pro/mc011460/3d-printer-filament-tpu-1-75

Toothed wheel (timing pulley) HTD 5M 20T



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 8

BUY plain bore + MODIFY

What it does

Same wheel as before, but Rev B's lanes are only 19-20 mm wide, so there is no room for a set-screw hub. The hinge ones are driven by the D-shaped shaft; the end ones spin on bearings on a shoulder screw.

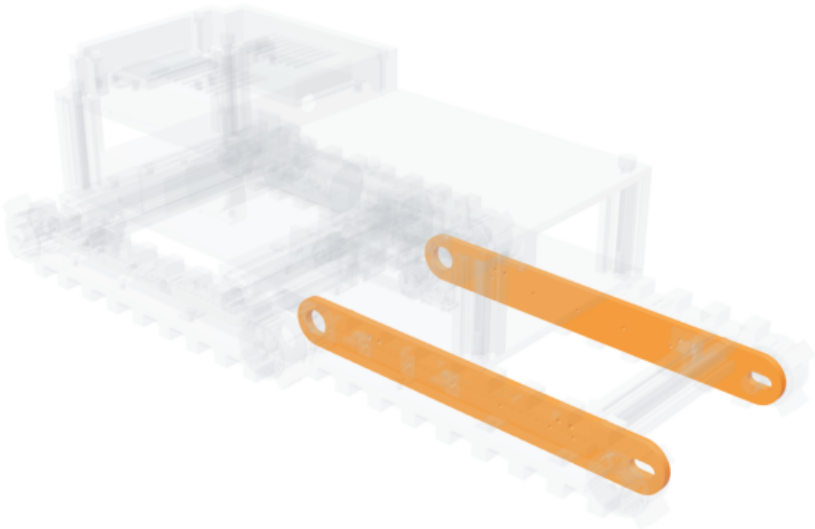
Where to get it

MISUMI SG, Timing Pulleys HTD 5M (link): 20 T, 15 mm belt, flanges both sides, PILOT (plain) bore. Then Professional Workshop: 4 bored to a D-shape Ø8 (hinge shaft, M1), 4 bored to Ø16 H7 for two 688 bearings each (end idlers, M2). Rev B sheet 21.

Link: MISUMI SG: HTD 5M timing pulleys

https://sg.misumi-ec.com/vona2/detail/110310532889/

Side plate, front segment (Rev B P1)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

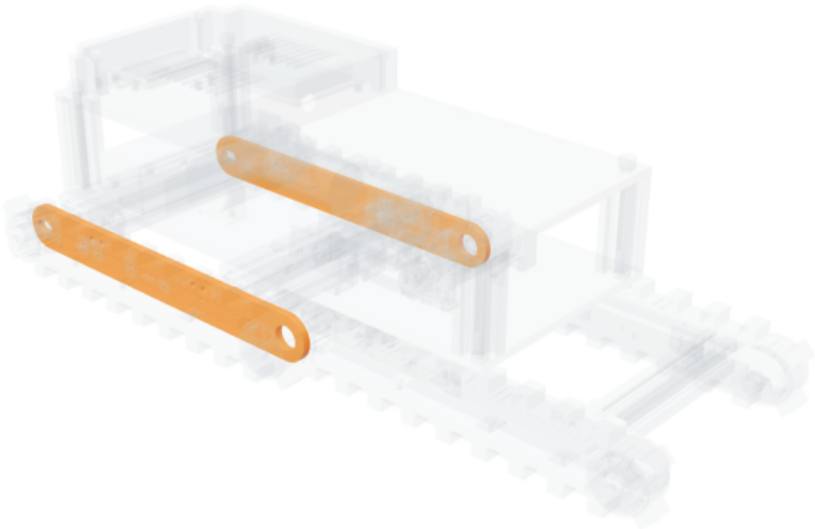
255 x 30.4 plate, inboard (z 55-60). Ø16 hole for the hinge bearing, a slot at the front end for the idler's shoulder screw (slide it to tension the track), tapped holes for the rail brackets and the 2020 bar brackets.

Where to get it

5 mm Al 6061, laser cut from DXF/P1 Side plate - FRONT segment.dxf, then drill and tap per Rev B sheet 4. Fab Lab laser does not cut aluminium: Professional Workshop or an outside laser shop on the vendor list.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Side plate, rear segment (Rev B P2)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

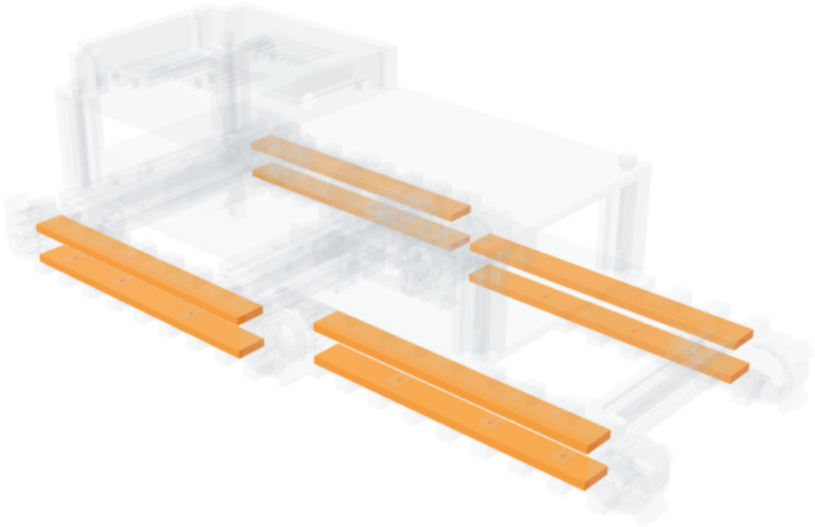
Outboard plate (z 80-85). The rear track runs outside it on a shoulder screw. Holes for the 3030 bar brackets and the rails.

Where to get it

As P1, from DXF/P2 Side plate - REAR segment.dxf, Rev B sheet 5.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Slide rails (Rev B P3-P6)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 8

BUY strip + drill

What it does

The strips the track's upper and lower runs slide on so the track cannot sag between the wheels. Does the job my printed rail did. My track's tooth tips ride on the lower one.

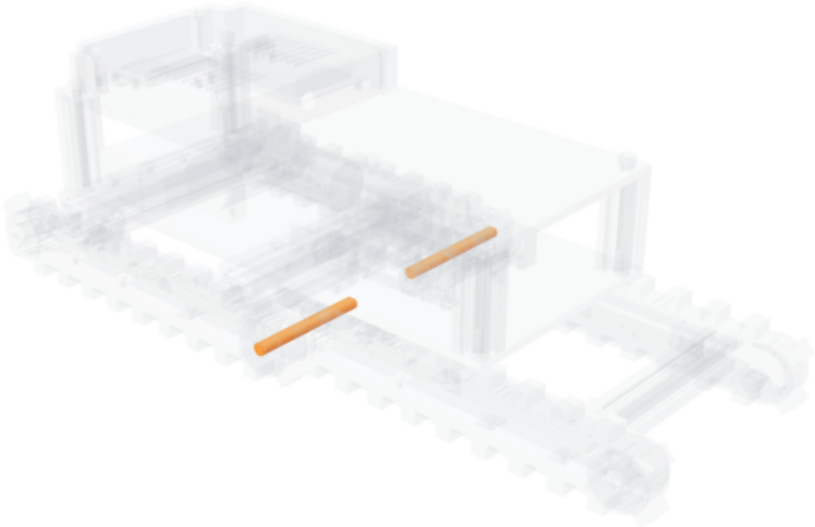
Where to get it

MISUMI SG UHMW-PE strip UPA-189-20-5 (link), 189 x 20 x 5, 3 pilot holes each, Rev B sheet 18. Held to the plates by RS PRO 427-991 angle brackets (RS SG) with M3 screws.

Link: MISUMI SG: plastic strips (UHMW)

https://sg.misumi-ec.com/vona2/mech/M1500000000/M1501000000/

Hinge shaft (Rev B P20 A + B)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 2 (or 1 if one motor)

BUY D-shaft, cut

What it does

Replaces my 230 mm middle axle. The drive pulleys sit on the D-flat, so no set screws. Runs in 688 bearings in both plates and in the P17 support.

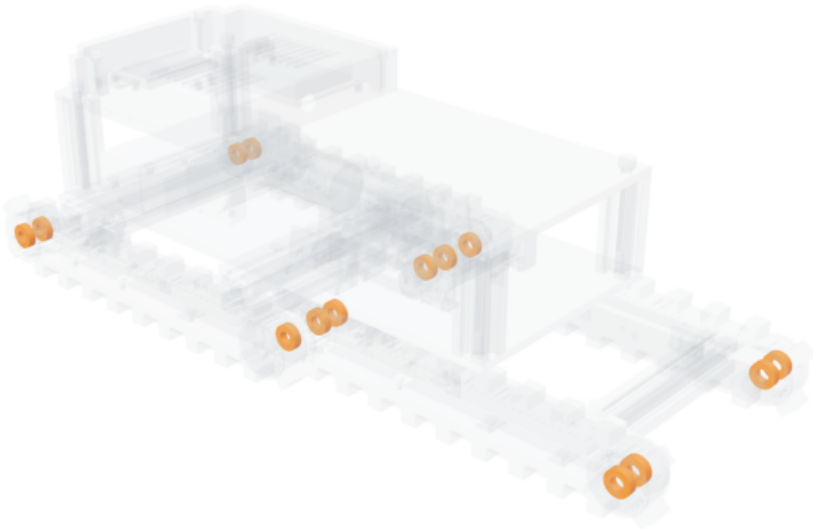
Where to get it

Ø8 h7 D-shaft, flat 7, 200 mm stock (MISUMI SG or RS SG linear shafts, link). Cut to 82.3 (A) and 77.3 (B); E-clip grooves per Rev B sheet 20. One motor per side -> two half shafts; one motor -> one 212 mm shaft.

Link: RS SG: linear shafts

https://sg.rs-online.com/web/c/bearings-seals/linear-motion/linear-shafts/

Ball bearing 688-2RS 8 x 16 x 5



Quantity: 14

BUY: RS Singapore

What it does

Replaces my 10 x 608ZZ. Smaller OD (16) because the plate is only 30 mm tall.

Where to get it

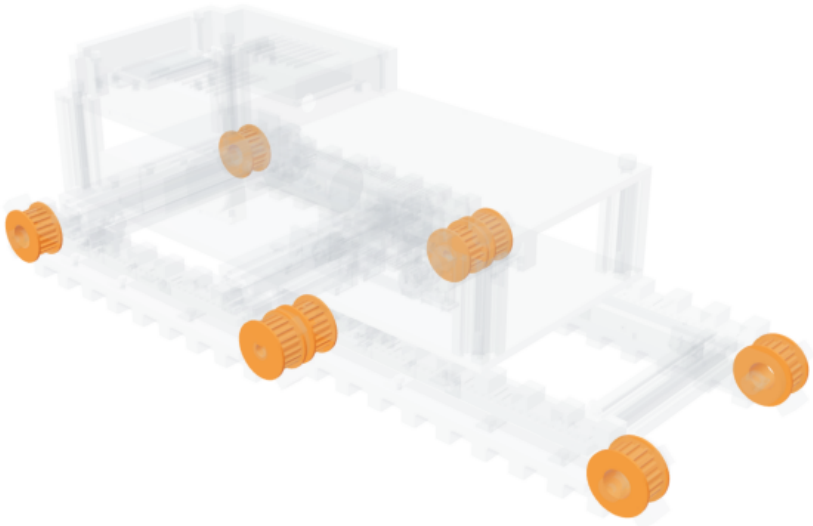
RS SG ball bearings (link): search 688-2RS. 4 in the plate hinge holes, 2 in the P17 supports, 8 inside the four idler pulleys.

Link: RS SG: ball bearings

https://sg.rs-online.com/web/c/bearings-seals/bearings/ball-bearings/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Idler stub axle: shoulder screw Ø8 x 25, M6



Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Quantity: 4

BUY: MISUMI SG or RS SG

What it does

Replaces my front and rear axles. One per plate end, through the slot; the end wheel spins on two 688s on its shoulder. Slide outwards in the slot and tighten the nut to tension the track.

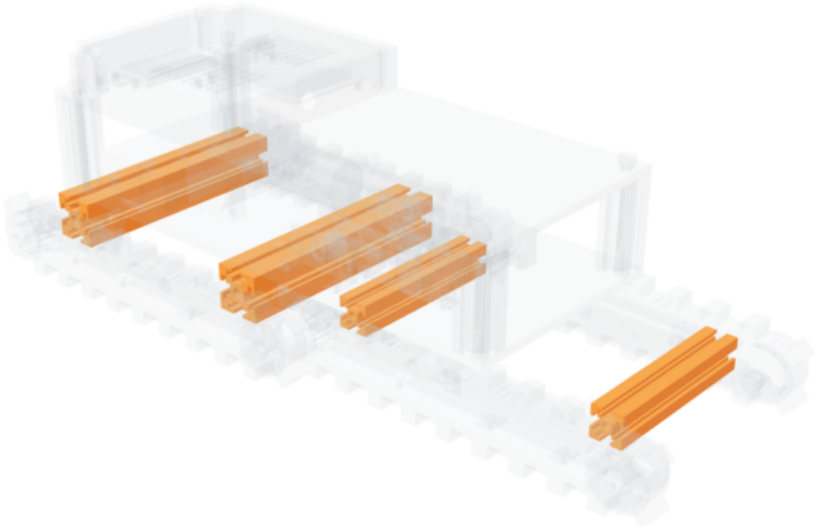
Where to get it

ISO 7379 8 x 25 shoulder screw (link), with 8 x 11 x 1 washer, 8 x 11 x 0.5 shim, 8 x 10 x 9 spacer, M6 washer and nyloc. Rev B sheet 3, idlers.

Link: RS SG: shoulder screws

https://sg.rs-online.com/web/c/fasteners-fixings/screws-bolts/shoulder-screws/

Cross bars: 2020 x 110 and 3030 x 160 (Rev B P7, P8)



Quantity: 2 + 2

BUY profile, cut

What it does

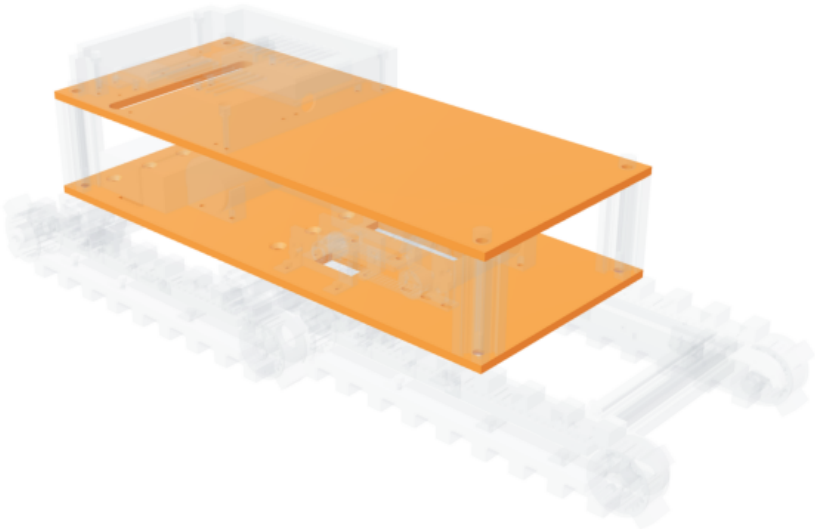
Replace my M4 rods and printed spacers: they hold the plates apart and carry the deck. The 3030 bars are also what the box/deck bolts to.

Where to get it

2020 B-type slot 6 (1 m) and 3030 slot 8 (0.5 m) from MISUMI SG or a local supplier; cut per Rev B sheet 19. Corner brackets P18/P19 from aluminium angle, T-nuts M5/M6.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.

Deck and tray (Rev B P12, P16) and everything on them



Quantity: 1 + 1

Nhien's parts

What it does

The 'box': 160 mm wide between the rear plates in this layout. Motors, drivers, battery and the platform mount sit here.

Where to get it

Al 5 mm laser cut, Rev B sheets 6-8; posts P15, controller cover P22, motors, electronics per the Rev B BOM.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model.